

Manual

Tool and Resource Monitoring WRM-WMO 8.1

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Contents

1	Resource Monitoring - Overview.....	4
2	Resource Overview	5
3	Resource history	13
4	Use of resources	19

1 Resource Monitoring - Overview

Purpose

The tool and resource monitoring packet contains functions for displaying the current state of resources, that is their status, their current storage location or order that is currently logged on. Furthermore, the resource history and use of resources functions are used for reporting purposes to evaluate work done with and on the resources. These evaluations, therefore, illustrate the maintenance history of the resources.

Integration

The entries in the overviews and reports are not only generated within resource management, but are sometimes created from other MES partial functions. The order postings, for example, come from the BDE.

Features

Resource overview with display of the current resource status, the current machine, the current operation and the locking flag incl. period.

Resource information such as master data, accumulated time and quantity totals and current status information are consolidated in a central function. Display of status, location and structural assignments.

Resource history reports: tabular listing of all resource events and uses that have accumulated.

Resource use report (downtimes). Cumulative display of the cycles and quantities entered for the resources with the ability to restrict the period and profile pattern for a period of time.

2 Resource Overview

Overview

Menu	Production facility management → Current information → Resource overview
Transaction code	resov
Function authorization	resov.*

The resource overview is the central application that provides an overview of the current resources managed in the system. The resource overview provides functions regarding status change, stock transfer and documentation of measures.

Purpose

You can use the *Resource overview* to show the current status of selected resources. The application shows all resources matching the conditions you specified in the selection parameters.

The list shows the specified resource master data and the current resource status. If the tool is active (logged on), the *Resource overview* shows the machine and the operation where the resource is logged on to. If the resource is locked, the overview shows the following:

- the reason of the lock,
- the time when the resource has been locked, and - if entered -
- the time until the resource will be locked.

Integration

You can use the master data function *Resource configuration* to create, change or delete resources.

Selection criteria

The application provides the following selection criteria:

Resource from ... to ...

This selection criterion refers to the resource. You can also use wildcards (placeholders *).

Short name

Short name of the resource. Only relevant for resources of type MNR.

Resource type

Type of resource.

Designation

Name of the resource.

Group

Workplace/machine group of the resource. Only relevant for resources of type MNR.

Cost center

Cost center of the resource.

Short name

Short name of the resource.

Resource family

Family the resource is assigned to.

Responsibility area

Responsibility area the resource is assigned to.

Storage location

Regular storage location of the resource.

Current storage location

Current storage location of the resource. The functions *Status change* and *Stock transfer* are used to book the resource to this storage location. If you log on a resource to a workplace, this workplace automatically becomes the current storage location of the resource.

Show deleted resources (as of WRM 8.2)

Use this field to specify if logically deleted resources are shown.

MD user fields

MD user fields 1- 6 of the resource. If you select a resource family in the selection panel, the application shows the field names according to the assigned user field definition.

Field descriptions

Most field descriptions are included in the application [Resource configuration](#). See the following tabs:

- General resource configuration
- User fields
- Comment
- Resource attributes
- Resource list

The additional tabs of the resource overview are described here:

"Status" tab**Active**

Specifies whether or not the resource is currently logged on to a workplace.

Deleted (as of WRM 8.2)

Specifies whether the resource displayed is logically deleted or not.

Current machine, current order

For active resources, this field specifies the workplace and the order where the resource is currently logged on.

Current storage location

Current storage location of the resource. The functions *Status change* and *Stock transfer* are used to book the resource to this storage location. If you log on a resource to a workplace, this workplace automatically becomes the current storage location of the resource.

Resource status, status since

Specifies the current status of the resource. Number and name are displayed and the time when the status was set.

New status, from, until

You can specify a future status that is set when the status changes. The status is not immediately set, but an expected future status is specified.

The HYDRA Shop Floor Scheduling can use this future status to show and monitor the resource assignment.

In the HYDRA Scheduler, a job "WRM resource control" is installed. This job regularly executes the script `./hyresctl.scr /STATUS` (by default every 30 minutes). The program called by the script, performs the following processing:

- If the time "New status from" is exceeded, the new status is set.
- If the time "New status until" is exceeded, then the status is set that includes the value "F" in field *Collection* in the [status configuration](#).

Technical note: Both status changes are internally performed via an asynchronous DDI call (DLG=RES_STATUS or DLG=RES_FREI).

"Use" tab**Target utilization****Cycles**

The field *Cycles* provides additional information. The cycles value defines how long the resource is to be used.

Runtime

The field *Runtime* provides additional information. It defines how long the resource is to be used.

Actual utilization

Cycles

The total number of cycles that have been recorded for the resource since initialization.

Runtime

The total run time that has been recorded for the resource since initialization.

Yield, scrap

Total of yield or scrap that has been recorded for the resource since initialization and posted to the relevant account (base quantity, primary quantity).

Run times since implementation, allocated to RPA

Account name (e.g. SUT, MUT)

Total number of hours that has been recorded for the resource and booked to this resource performance account since initialization.

Toolbar

Resource tab



File – Show file (function authorization: mdres.doc)

Opens the file view – only available for document resources, which are configured as file-based resources without DNC processing in the *Resource type*. And only available if the relevant license and function authorization are available.



Go to – Resource configuration (function authorization: mdres)

Opens the application *Resource configuration*. The selected resource is passed as default value.



Go to – Resource list (function authorization: mdrbom)

Opens the *Resource list* application. The selected resource is entered as default value for the higher-level resource.



Go to – Resource history (function authorization: reshi)

Opens the application *Resource history*. The selected resource is passed as default value.



Functions – Measure

Opens the *Measures* application.



Functions – Status change

Opens the dialog to change a resource status. The following input fields are available:

Status: Status that is set according to configuration.

Material buffer: With the status change, the resource can be relocated and transferred to another storage location.

Set immediately:

- The current date is preassigned to the field *from* and cannot be changed.
- The field *till* is empty; you can enter a date here.

Set for period:

- The current date is preassigned to the field *from* and can be changed.
- The field *till* is empty; you can enter a date here.

If no date is entered in field *till*, the system automatically sets the date to 31-DEC-9999.

The checkbox *Including subordinate resources* is not relevant and reserved for future extensions.

Comment: The comment entered can be displayed in the *Resource history*.



Functions – Release of resource

Opens the dialog to release a resource. The checkbox *Including subordinate resources* is not relevant and reserved for future extensions.



Functions – Stock transfer

Opens the dialog to transfer/relocate a resource.



Generate order (function authorization: resovgenorder)

Generates a maintenance order for the selected data row. (The function is only available with function package WRM-IHA).



Document management (function authorization: resovdoc)

Click this button to call the [Document management](#).

DNC tab

The tab is only available, if you select a DNC resource. These are resources configured with *DNC processing* in the *Resource type*.

**File – Export (function authorization: mdres.export)**

Exports the file specified for the resource. You use the file explorer to specify the target file.

**File - Import (function authorization: mdres.import)**

Imports the file specified for the resource. You use the file explorer to specify the source file.

**File - Viewer (function authorization: mdres.vis)**

Opens the file specified for the resource using the defined viewer program.

**File - Editor (function authorization: mdres.editor)**

Opens the file specified for the resource for editing using the defined editing program.

**File – Comparison editor (function authorization: mdres.diff)**

Opens the comparison editor for the selected resource or resources. See below for further information.

**Go to – Resource configuration (function authorization: mdres)**

Opens the application *Resource configuration*. The selected resource is passed as default value.

**Go to – Resource list (function authorization: mdrbom)**

Opens the *Resource list* application. The selected resource is entered as default value for the higher-level resource.

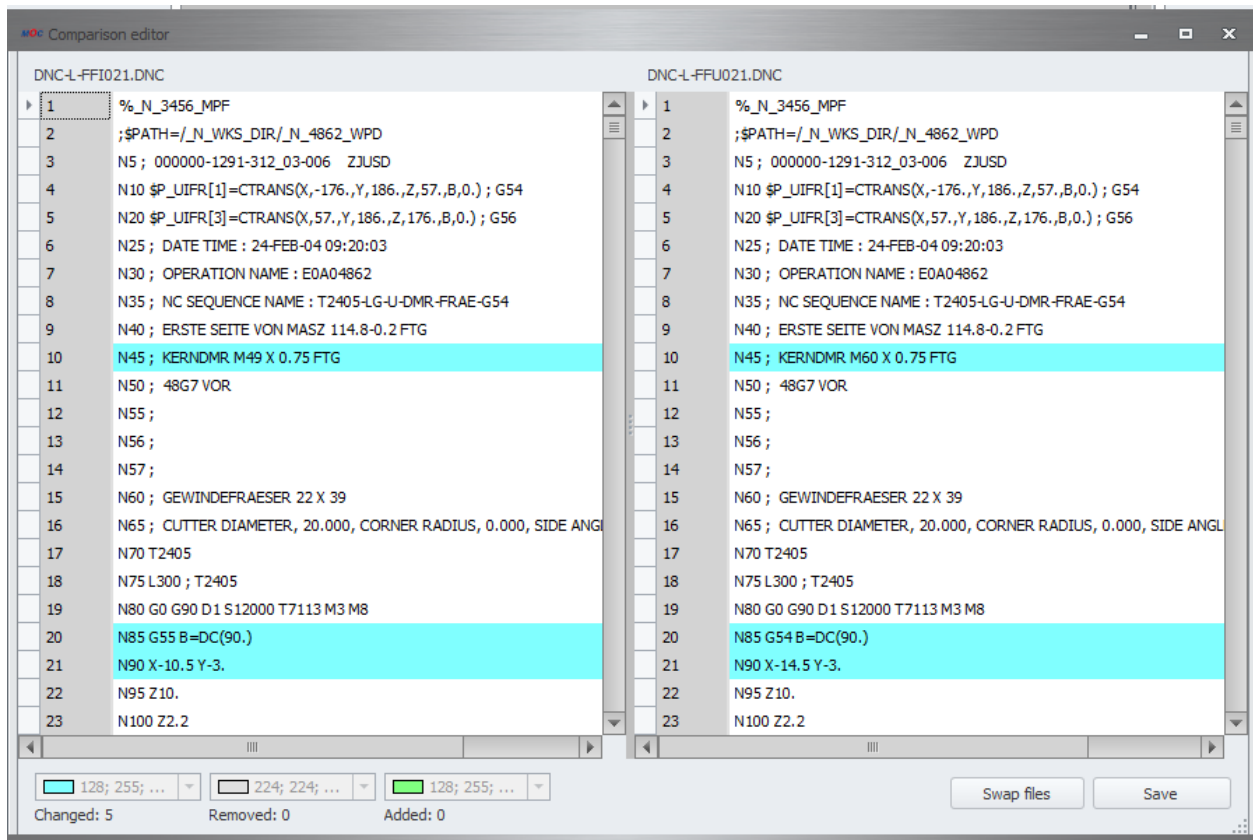
**Functions – Status change**

Opens the dialog to change a resource status.

**Functions – Release of resource**

Opens the dialog to release a resource.

How to use the comparison editor



The comparison editor compares the files attached to the DNC resources. Two operation modes are available:

Selection of one resource:

The editor shows the released resource and the optimized version of the resource for comparison. You can change the file displayed on the right-hand side of the editor. Once you have made the changes, the comparison editor transfers these changes to the system, like the simple editor. You can only use this mode for DNC types with the file processing type "optimized".

Selection of two resources:

If you select two resources before you open the comparison editor, the editor compares the two selected resources. You can select the file type. You can change the file displayed on the right-hand side of the editor. Once you have made the changes, the comparison editor transfers these changes to the system, like the simple editor.

Click the relevant buttons or use the context menu (right-click) to start the functions of the comparison editor:

- Reject: Rejects the difference identified (on the right). Accepts the value from the left file. The editor does no longer highlight the difference.
- Keep: Accepts the difference identified (on the right). The editor does no longer highlight the difference.
- Next difference: Goes to the next difference.
- Insert: Inserts a row at the current position.
- You can always change the contents of a row. Click the row and enter a value. Press ESC to quit the row without changes. The editor then highlights the row as "changed".
- Swap windows: Click this button to swap the windows. This function is necessary if you compare two resources. The place where a resource is displayed results from the display order in the table; the system does not know, which resource must be changed. In mode *Selection of one resource*, this button is not available because here you can only change the optimized program version.
- Save: Saves the changes made to the file on the left-hand side.

3 Resource history

Overview

Menu	Production facility management → Resource analysis → Resource history
Transaction code	reshi
Function authorization	reshi

The "resource history" provides an overview of what happened to a resource in the past. Therefore, you can trace activities that are relevant to quality ("log book"). The application maps the resource's "life cycle".

Purpose

In this application the system documents all actions/events relating to the object "Resource". The resource history documents the actions/events resulting from the resource management and the use of resources in BDE orders. BDE orders can be production orders and maintenance orders. The system documents the following:

- Status changes (resource status changes)
- Stock transfers
- Measures/ comments
- Exceeded maintenance cycles
- Maintenance reset
- With DNC: Upload and download
- Order logons
- Order logoffs

Whatever has been done to and with the resources is documented and available at any time. The application also includes a print function in the electronic resource book.

Selection criteria

The application provides the following selection criteria:

Resource type

This selection criterion refers to the resource type. You can also use wildcards (placeholders *).

Date from / to

Use the date selection to restrict the period of time for the data you want to evaluate.

Both times respectively refer to the start or end of the period specified above.

Resource

This selection criterion refers to the resource number. You can also use wildcards (placeholders *).

Events

Use this selection criterion to select specific events. The events you can select depend on the product groups you have licensed and the events that are actually recorded.

Workplace

Enter a workplace to select the events that have been posted in relation to this workplace. This mainly affects the following events:

- Machine status
- Beginning of status / end of status
- Production lock
- Operation postings
- Personnel postings
- Target value changes
- Resource postings
- BDE comments
- DNC upload/ download

Family

This selection criterion refers to the resource family the resource is assigned to.

Status

This selection criterion refers to the status of the resource. The system selects the events of the following types for which this resource status was set in the period entered:

- Resource status
- Release of resources
- Maintenance cycle exceeded
- Maintenance reset
- DNC upload
- DNC download

Cost center (workplace)

This selection criterion refers to the cost center stored in the machine and/or workplace master data. The application shows all machines and/or workplaces assigned to the selected cost center. You can also use wildcards.

Designation (name)

This selection criterion refers to the resource name as defined in the workplace/resource configuration.

Cost center (resource)

This selection criterion refers to the cost center of the resource as defined in the workplace/resource configuration.

Article

This selection criterion refers to the article number of the operation that was recorded with the event.

The system logs the operation with the following events:

- Logging on/off, interrupting operations (for resources of the type MNR)
- Logging on/off staff (for resources of the type MNR)
- BDE comments (for resources of the type MNR)
- Changing the target cycle (for resources of the type MNR)
- Changing the partitioning (for resources of the type MNR)
- Logging on/off resources (for resources of the type <> MNR)

MES order number

This selection criterion refers to the MES order number of the operation that was recorded with the event.

Order

This selection criterion refers to the order number of the operation that was recorded with the event.

Reporting person

This selection criterion refers to the personnel number of the employee who carried out the posting that initiated the event.

The system logs the *person* for the following events, if the person was entered with the posting:

- Logging on/off, interrupting operations (for resources of the type MNR)
- Logging on/off staff (for resources of the type MNR)
- BDE comments (for resources of the type MNR)
- Changing the workplace/machine status (for resources of the type MNR)
- Changing the target cycle (for resources of the type MNR)
- Changing the partitioning (for resources of the type MNR)
- Logging on/off resources (for resources of the type <> MNR)

Field descriptions



Subject to the logged events, the system only populates specific fields.

The "Duration" field, for example, is populated in case of a machine status change. But if the resource status changes, the system enters the value zero.

The fields are classified into the areas:

Event

Includes information on the event.

Resource

Provides information on the currently selected resources. The event usually refers to the resource mentioned in the subsection "Resource master data".



If an event only refers to a machine, the relevant machine data is entered in the "Resource master data" section and in the "Workplace master data" section.

If a DNC resource is either uploaded or downloaded, the DNC resource is entered in the "Resource master data" section. The "workplace master data" section shows the machine from which data was uploaded or downloaded.

If a resource is changed (e.g. a resource is logged on/off via a terminal), the resource is entered in the "Resource master data" section. The "Workplace master data" section includes the relevant machine.

Person

Information about the person or user performing the posting.



If you reset a maintenance activity via a terminal, the application also shows the terminal user number in the "Modified by" field. If you reset a maintenance via the MOC, the field includes the user (MOC user).

The other columns depend on the event.

Resource status

Information on the set resource status of the status event.

Order

Information on the operation that was currently processed at the time of the event (event timestamp)

Maintenance

Information on the *maintenance* event.

Measure

Information on the *measure* event

Upload/ download

Information on the DNC event *upload* and/or *download*.

Note: Two events are documented with the upload, as the upload is processed in two stages. Dialog ID "N": upload has started. "F": file has been saved in the target folder. The system only makes one entry for downloads. Only the file transfer is documented as a relevant event.

Toolbar



Resource overview

You can directly open the application "resource overview" by clicking the icon.



Generate order (reshigenorder)

Use the "generate order" function to create orders from work plans based on the specified configuration.



Document management (reshidoc)

This button is only enabled if you select a "Maintenance reset" event. The [Document management](#) opens.



Set the measure to "measure completed" (resmeasfin)

Use this function to identify an unfinished measure as being "completed". Once you have requested the function, a prompt pops up where you have to confirm your input. Once you have confirmed the dialog, the system sets the measure to "done" (column "done"/"settled": ) and documents the user (column "done by").



Resource book (reshi.book)

The resource book is a report providing the information of the resource history in a formatted form. The aim of this function is to make data printable. This way, the data is also available as hard copy, e.g. for QA documentation purposes. You may choose to have the data grouped by day for the daily report or by week for the weekly report.

The cover sheet shows master data information of the displayed and selected resources.

The presentation varies depending on the event.

4 Use of resources

Overview

Menu	Production facility management → Resource analysis → Use of resources
Transaction code	Resemp
Function authorization	Resemp

Purpose

Depending on the selection made, this evaluation displays the actual values of resources to which postings were made for time, quantities and cycles/strokes/shots. You can select the time period you want to consider. You may specify the resources you wish to view in the upper part of the selection pane. The selection is based on the logon time of the posting. The time for data selection may vary depending on the extent of the selection period.

In addition to resource-related master data, the application provides the following data:

- Cycles/strokes/shots posted since implementation or within a certain period of time (depending on selection)
- Quantity posted since implementation or within a certain period of time (depending on selection)
- Downtimes times posted since implementation or within a certain period of time (depending on selection), totaled and divided up according to resource performance accounts (RPAs)

The application also pulls information from postings already transferred into archive tables.



The evaluation only integrates those resources that are assigned to the "consider in evaluations" option in the "configuration" category of the resource stock. The option "Post to resource" needs to be checked in order for quantities, cycles and times to be posted to resources. Posting takes place when the operation to which the resource is assigned is logged on.

File-based resources and resources with DNC processing can be neither logged in nor posted.



After changing or adding data records, you have to refresh the data manually to view the changed/added data (click the "request data button").

Selection criteria

The application provides the following selection criteria:

Resource

Selected resource.

Resource type

You can filter data by the resource type.

Family

You can filter data by the resource family.

Workplace

Machine where the resource was logged on.

Order

You can filter data by the logged in order.

Article

You can filter data by the article of the logged in order.

Field descriptions**General, master data**

Shows the resource master data.

Order, article machine/workplace

Shows order data.

Date

Date of the included postings; including separate fields for year, month and day. You can use the fields to group data:

- Date/time specification
- Year
- Month
- Day

Quantities

Quantity totals of the postings.

RPA

Account totals of the postings.

Toolbar

Main page tab



Insert

Opens the dialog for adding data.



Copy

Opens the dialog for copying data.



Edit

Opens the dialog for editing data.



Delete

Deletes the selected entry.



Authorize

Not relevant at the moment. For later use.